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Bug Tracking Project — Submission (BugReport)

Problem Statement #17 — Healthcare & Telemedicine: *Vaccination Cohort & Dose Scheduling System*

A public health administration platform to organize demographic vaccination rollouts, enforce minimum interval days between doses, schedule center slots, and generate verifiable QR digital vaccination certificates.

Actors: Citizen Registrant, Vaccination Officer

Overview

A separate Bug Tracker project was created in Jira to log defects found while validating the 5 FRs and 2 NFRs of BPS#17 (Vaccination Cohort & Dose Scheduling System). Each bug below traces back to the requirement it violates.

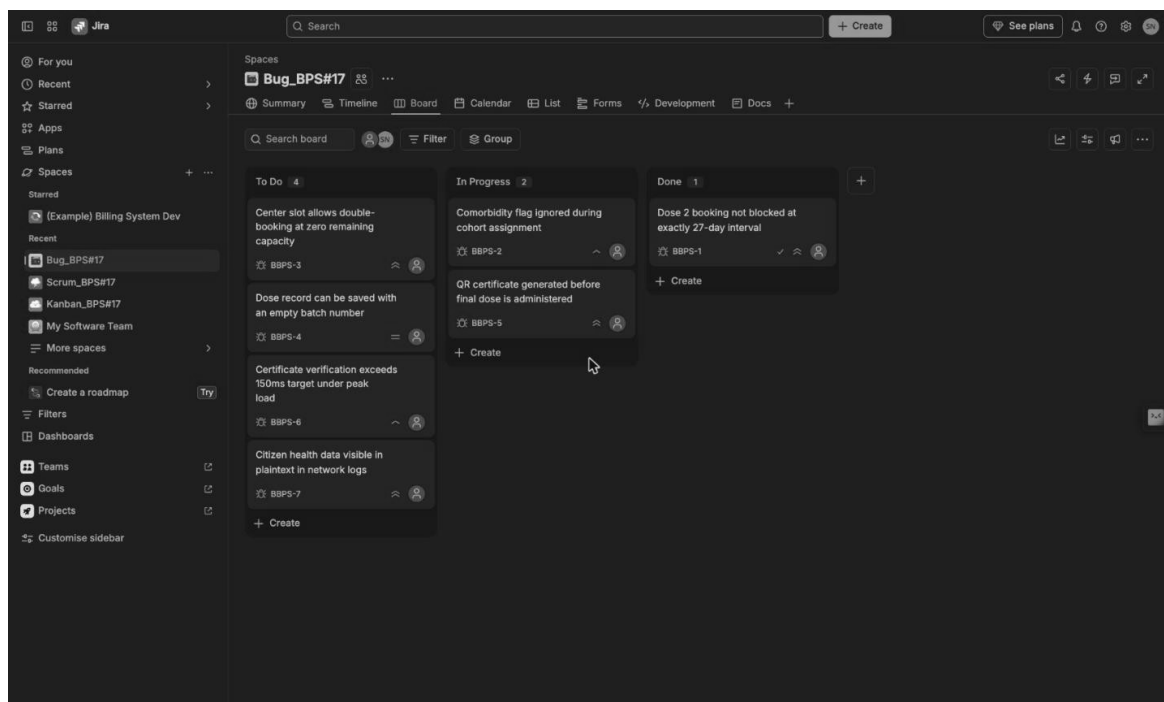


Figure 1. Bug_BPS#17 board (BBPS project, Kanban template) after all 7 bugs were logged and triaged: To Do holds BUG-003, BUG-004, BUG-006, BUG-007 (4 items); In Progress holds BUG-002 and BUG-005 (2 items); Done holds BUG-001 (1 item, resolved and closed).

Bug Summary

Bug ID	Summary	Related Req.	Severity	Status
BUG-001	Dose 2 booking not blocked at exactly 27-day interval	FR-001	Critical	Done
BUG-002	Comorbidity flag ignored during cohort assignment	FR-002	High	In Progress
BUG-003	Center slot allows double-booking at zero remaining capacity	FR-003	Critical	Open
BUG-004	Dose record can be saved with an empty batch number	FR-004	Medium	Open

Bug ID	Summary	Related Req.	Severity	Status
BUG-005	QR certificate generated before final dose is administered	FR-005	Critical	In Progress
BUG-006	Certificate verification exceeds 150ms target under peak load	NFR-001	High	Open
BUG-007	Citizen health data visible in plaintext in network logs	NFR-002	Critical	Open

Bug Details

BUG-001: Dose 2 booking not blocked at exactly 27-day interval

Related Requirement	Severity	Board Priority	Status
FR-001	Critical	Highest	Done

Steps to reproduce: 1) Complete Dose 1. 2) Attempt to book Dose 2 on day 27. 3) Booking succeeds instead of being blocked.

Likely root cause: Off-by-one error in the interval-comparison logic (used <= instead of <).

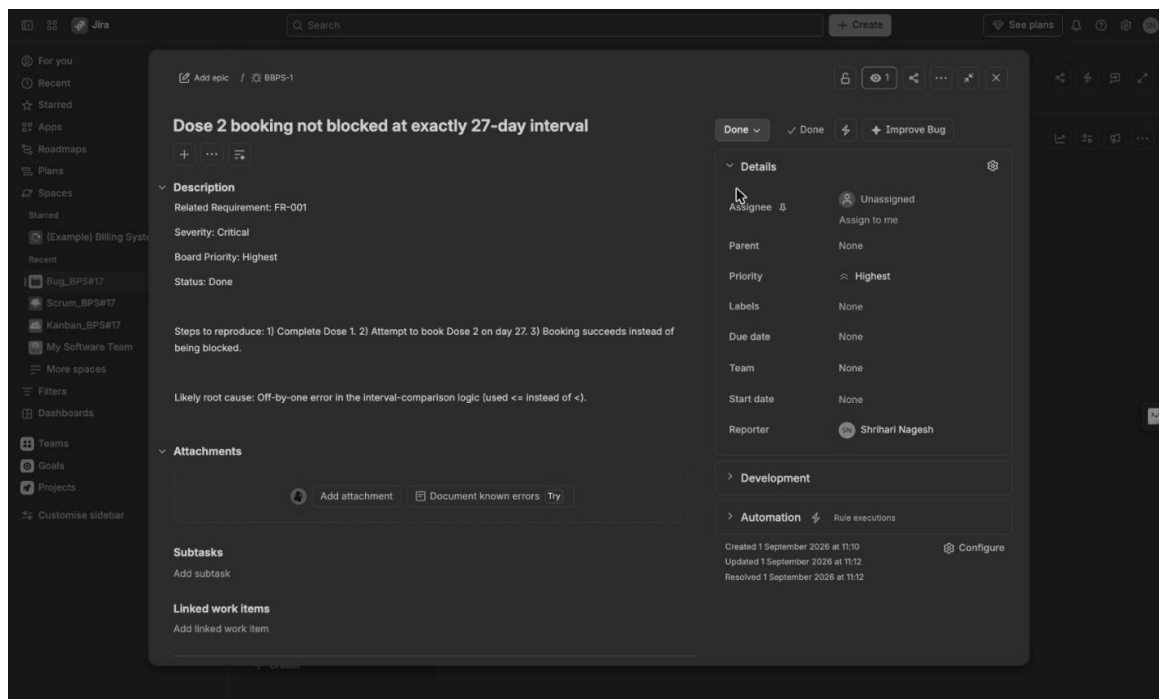


Figure 2. BUG-001 (BBPS-1) detail view: "Dose 2 booking not blocked at exactly 27-day interval", Priority Highest, Status Done, with the description recording Related Requirement FR-001, Severity Critical, Board Priority Highest, steps to reproduce, and the off-by-one root cause.

BUG-002: Comorbidity flag ignored during cohort assignment

Related Requirement	Severity	Board Priority	Status
FR-002	High	High	In Progress

Steps to reproduce: 1) Register with a comorbidity flag set to true. 2) Check assigned cohort. 3) Citizen is placed in the general (non-priority) cohort.

Likely root cause: Cohort rule engine only reads the age field; the comorbidity flag is not passed into the rule evaluation.

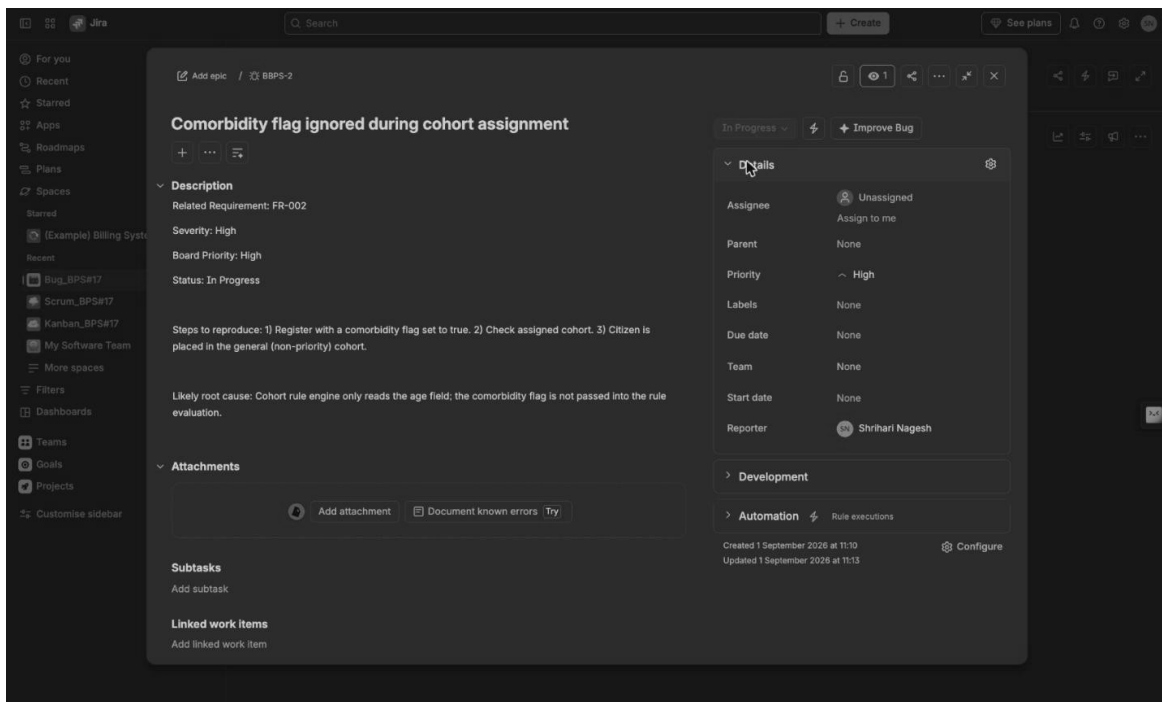


Figure 3. BUG-002 (BBPS-2) detail view: "Comorbidity flag ignored during cohort assignment", Priority High, Status In Progress, with the description recording Related Requirement FR-002, Severity High, Board Priority High, steps to reproduce, and the root cause (comorbidity flag not passed into the rule engine).

BUG-003: Center slot allows double-booking at zero remaining capacity

Related Requirement	Severity	Board Priority	Status
FR-003	Critical	Highest	Open

Steps to reproduce: 1) Reduce a center's remaining capacity to 1. 2) Two citizens submit a booking for the same slot simultaneously. 3) Both bookings are confirmed.

Likely root cause: Slot capacity check and decrement are not wrapped in a single atomic transaction (race condition).

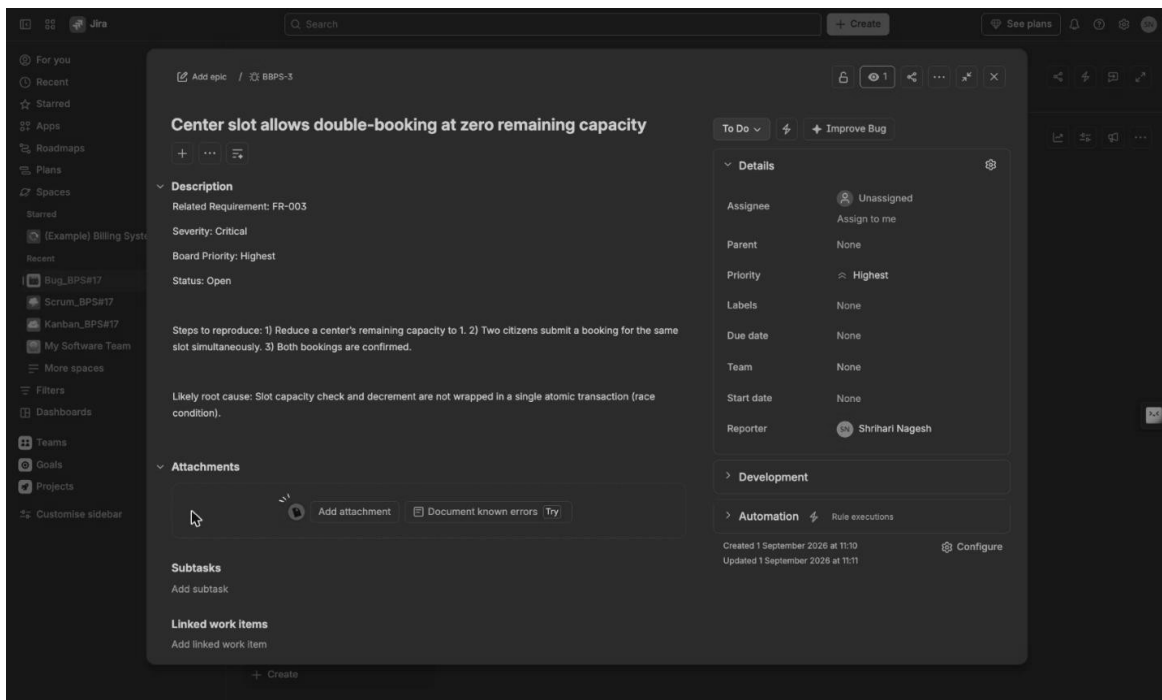


Figure 4. BUG-003 (BBPS-3) detail view: "Center slot allows double-booking at zero remaining capacity", Priority Highest, Status To Do (Open), with the description recording Related Requirement FR-003, Severity Critical, Board Priority Highest, steps to reproduce, and the race-condition root cause.

BUG-004: Dose record can be saved with an empty batch number

Related Requirement	Severity	Board Priority	Status
FR-004	Medium	Medium	Open

Steps to reproduce: 1) Officer opens check-in form. 2) Leaves the batch/lot number field blank. 3) Saves the record successfully.

Likely root cause: Missing required-field validation on the batch-number input.

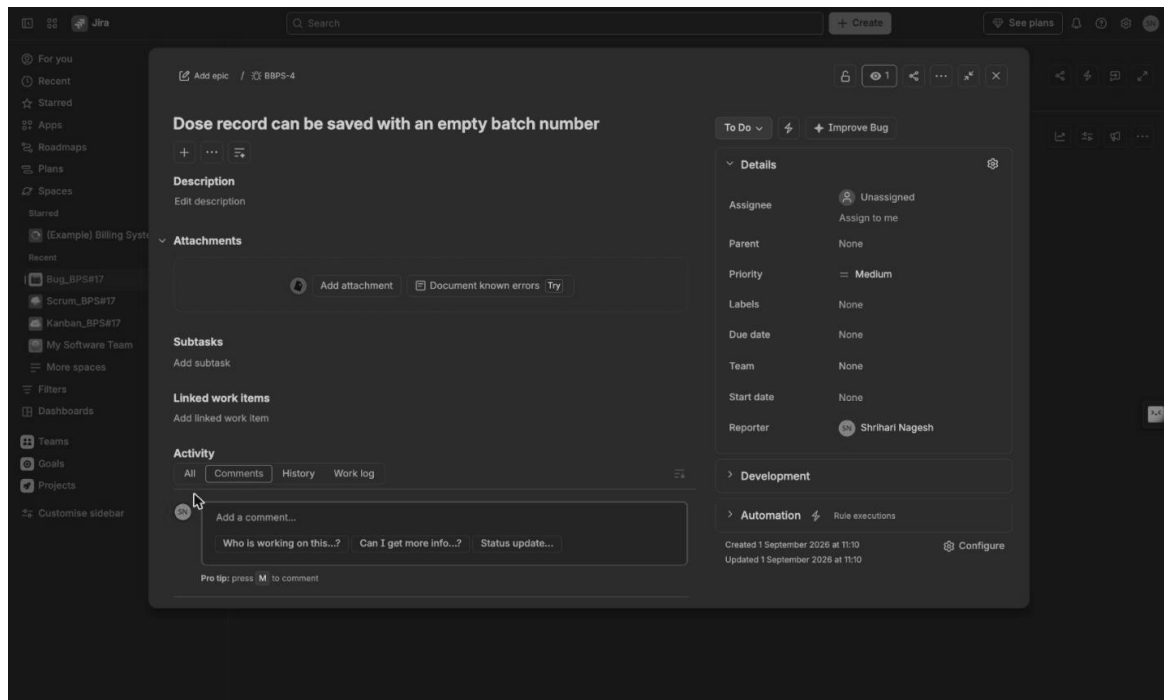


Figure 5. BUG-004 (BBPS-4) detail view: "Dose record can be saved with an empty batch number", Priority Medium, Status To Do (Open), with the description recording Related Requirement FR-004, Severity Medium, Board Priority Medium, steps to reproduce, and the missing-validation root cause.

BUG-005: QR certificate generated before final dose is administered

Related Requirement	Severity	Board Priority	Status
FR-005	Critical	Highest	In Progress

Steps to reproduce: 1) Citizen completes Dose 1 of a 2-dose vaccine. 2) Navigate to the certificate screen. 3) A certificate is generated and downloadable.

Likely root cause: Certificate trigger checks only "at least one dose recorded" instead of "dose count == required doses."

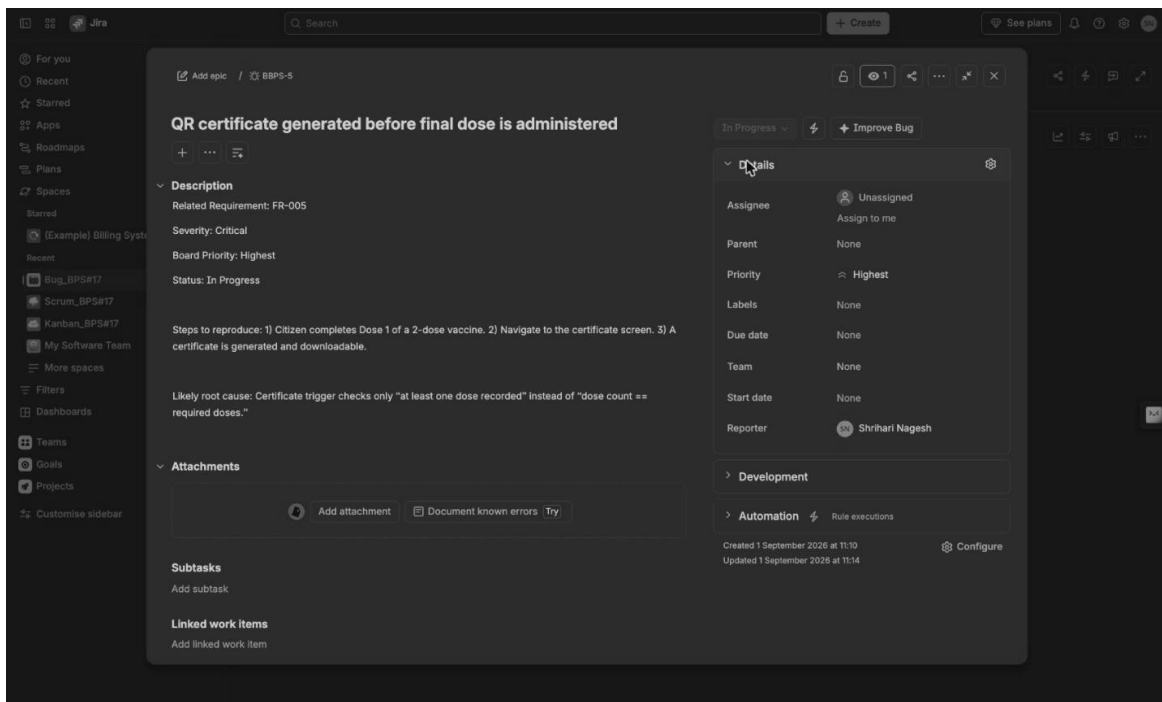


Figure 6. BUG-005 (BBPS-5) detail view: "QR certificate generated before final dose is administered", Priority Highest, Status In Progress, with the description recording Related Requirement FR-005, Severity Critical, Board Priority Highest, steps to reproduce, and the certificate-trigger root cause.

BUG-006: Certificate verification exceeds 150ms target under peak load

Related Requirement	Severity	Board Priority	Status
NFR-001	High	High	Open

Steps to reproduce: 1) Run the load test script at 500 concurrent verification requests. 2) Observe p95 latency of ~400ms.

Likely root cause: Public-key lookups hit the database on every request instead of using an in-memory cache.

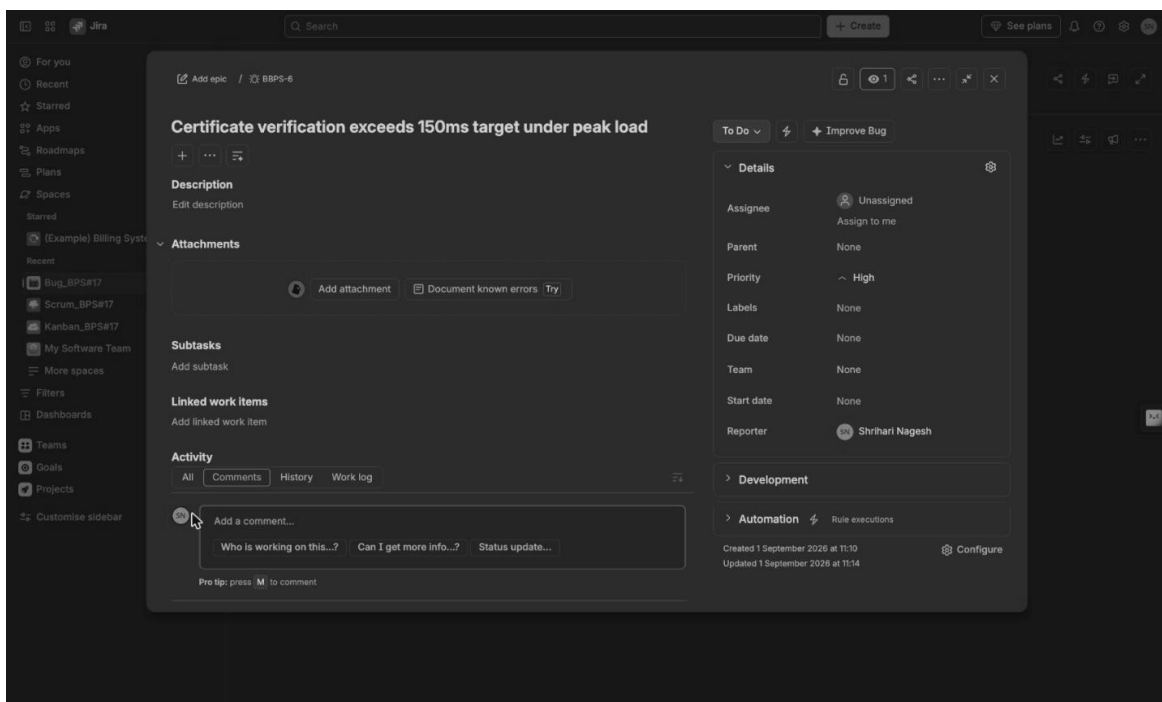


Figure 7. BUG-006 (BBPS-6) detail view: "Certificate verification exceeds 150ms target under peak load", Priority High, Status To Do (Open), with the description recording Related Requirement NFR-001, Severity High, Board Priority High, steps to reproduce, and the public-key-caching root cause.

BUG-007: Citizen health data visible in plaintext in network logs

Related Requirement	Severity	Board Priority	Status
NFR-002	Critical	Highest	Open

Steps to reproduce: 1) Open browser dev tools while checking a citizen's vaccination history. 2) Inspect the network response. 3) Response body contains unencrypted health data.

Likely root cause: One internal reporting endpoint was left off the TLS-termination allow-list and serves over plain HTTP.

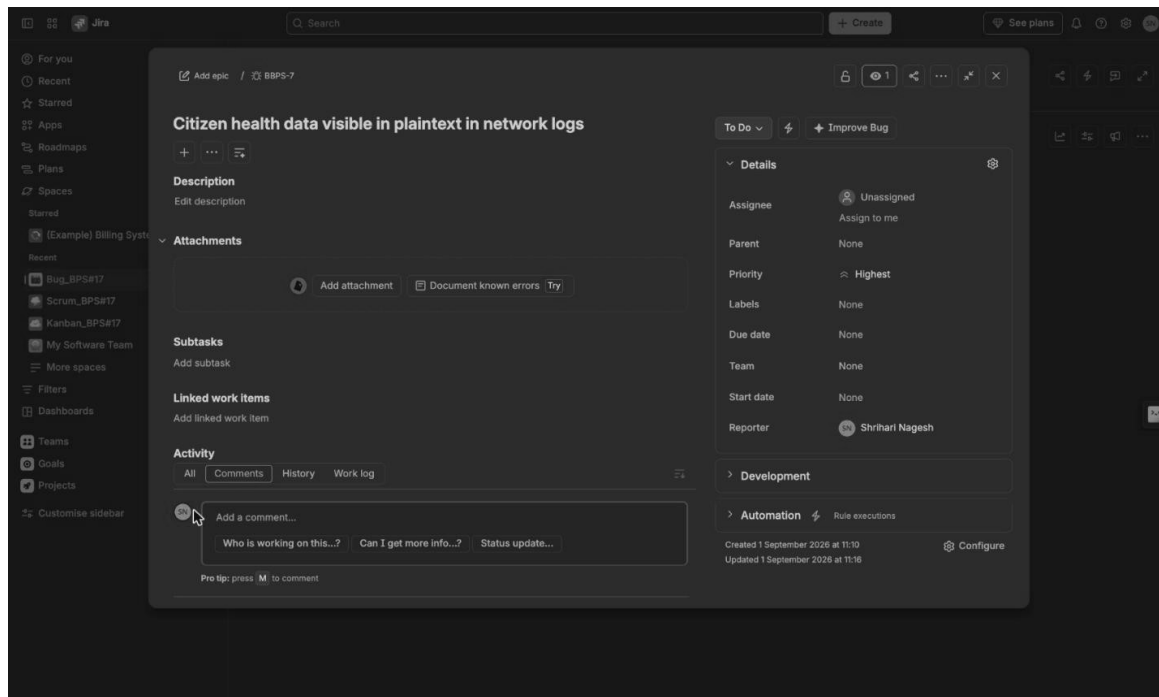


Figure 8. BUG-007 (BBPS-7) detail view: "Citizen health data visible in plaintext in network logs", Priority Highest, Status To Do (Open), with the description recording Related Requirement NFR-002, Severity Critical, Board Priority Highest, steps to reproduce, and the TLS-allow-list root cause.